DEVELOPMENT OF A SMART INSULIN PUMP FOR DIABETIC PATIENTS

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CERTIFICATION

This is to certify that this report on "**A SMART INSULIN PUMP FOR DIABETIC PATIENTS**" submitted to the Department of Biomedical Engineering, University of Lagos, for the award of the Bachelor of Science (Honors) in Biomedical Engineering, is a record of the work carried out by **ONYEJIAKA MADUABUCHI KYRIAN** with matriculation number **170410008** and **LAWAL AYOBAMI SALIM** with matric number **170410023**

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DEDICATION

This report is dedicated to our loving parents, Mr. and Mrs. Lawal, Mrs. Onyejiaka and Engr. Paul whose unwavering support and encouragement have been our greatest source of strength. We also dedicate this work to Engr. Ugochi , whose guidance and wisdom has inspired us all through this journey and also a special shout out to our friends and coursemates. Above all, we thank God for His blessings and guidance.

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ABSTRACT

This project focuses on the development of a smart insulin pump system designed to monitor and manage glucose levels in diabetic patients. Diabetes remains a significant global health issue, with an estimated 537 million adults affected worldwide. The project aims to address this by integrating a Near-Infrared (NIR) glucose monitoring system with a mobile application and an automatic insulin pump. The NIR method allows for non-invasive glucose measurement, which is then processed and displayed in real-time via a user-friendly mobile interface. The system is also equipped with an emergency alert feature and can trigger the insulin pump automatically when glucose levels reach critical thresholds. The study involved extensive testing of the NIR sensor's accuracy and reliability, the design of a stepper motor-driven insulin pump, and the development of the corresponding mobile application. Results indicate that the NIR sensor provides accurate glucose readings, and the automated insulin delivery system responds effectively to glucose fluctuations, contributing to better diabetes management. This work contributes to the growing field of diabetes care technology by offering a novel, non-invasive solution that improves patient autonomy and safety. The smart insulin pump system represents a significant advancement in personalized healthcare, with potential applications beyond diabetes management. In conclusion, the integration of NIR technology with real-time data processing and automated insulin delivery could revolutionize the way diabetes is managed, providing a more efficient and patient-friendly approach.

KEYWORDS: Near Infrared sensor, continuous glucose monitoring, telemedicine , insulin pump

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ABBREVIATIONS

NIR: Near Infrared Radiation

MIR: Mid Infrared Radiation

FIR: Far Infrared Radiation

IC: Integrated Circiut

IoT: Internet of Things

CNC: Computer numerical control

CGM: Continuous Glucose Monitoring

CSII - Continuous Subcutaneous Insulin Infusion

HbA1c - Hemoglobin A1c

LED - Light Emitting Diode

NIRS - Near-Infrared Spectroscopy

nm - Nanometers

IoT: Internet of Things

UI: User Interface

ESP32: Espressif Systems' ESP32

BLE: Bluetooth Low Energy

JSX: JavaScript XML

UUID: Universally Unique Identifier

BLE: Bluetooth Low Energy

GIT: Git Version Control

C++: C Plus Plus

IDE: Integrated Development Environment

GL: Glucose Levels

UI: User Interface

CVS: Continuous Visualisation System

EMF: Electromotive Force

CAD: Computer-Aided Design

AI: Artificial Intelligence

3D: Three-Dimension

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Diabetes mellitus, a chronic metabolic disorder characterized by elevated blood glucose levels, affects an estimated 537 million adults worldwide, with projections suggesting this number could rise to 783 million by 2045 (International Diabetes Federation, 2021). Effective management of diabetes hinges on accurate and timely monitoring of blood glucose levels, coupled with appropriate insulin administration. Traditional glucose monitoring methods, typically involving frequent finger pricks for blood sampling, can be both inconvenient and painful for patients, potentially leading to reduced compliance with recommended testing regimens (Heinemann et al., 2018).

In recent years, continuous glucose monitoring (CGM) systems have emerged as a promising alternative to traditional methods, offering real-time glucose data and improved insights into glucose trends (Rodbard, 2017). Non-invasive glucose monitoring techniques, such as those using Near Infrared radiation, have the potential to alleviate these challenges (Liu et al., 2020).

However, many current CGM systems still rely on minimally invasive sensors inserted under the skin, which can cause discomfort and require periodic replacement.

This research project aims to develop an innovative smart insulin pump system that incorporates truly non-invasive continuous glucose monitoring using near infrared (NIR) sensors. The smart insulin pump will utilize Near Infrared radiation to detect glucose levels and a custom-made insulin pump to administer insulin when the glucose level is low (Sunny & Kumar, 2018).

Near Infrared radiation is a form of electromagnetic radiation that can penetrate the skin and interact with glucose molecules, allowing for the measurement of glucose levels without the need for invasive procedures (Gonzales et al., 2019).

Near infrared spectroscopy has shown potential for non-invasive glucose monitoring due to its ability to penetrate human tissue and interact with glucose molecules (Yadav et al., 2015). The proposed system seeks to leverage NIR technology to overcome limitations of current glucose monitoring methods by offering:

1. Non-invasive monitoring, significantly reducing patient discomfort and potentially improving adherence to testing protocols.
2. Continuous data collection, providing a more comprehensive view of glucose trends and facilitating more informed treatment decisions.
3. Potential for improved accuracy compared to traditional intermittent testing, particularly in capturing rapid glucose fluctuations.
4. Integration with insulin delivery systems for more precise and automated diabetes management.

The development of such a system presents several technical challenges, including:

1. Optimizing NIR sensor design for glucose specificity and sensitivity in the presence of interfering substances (Uwadaira & Ikehata, 2018).
2. Developing robust algorithms for data processing, noise reduction, and accurate glucose prediction from NIR spectral data (Chen et al., 2018).
3. Integrating the NIR sensing system with insulin delivery mechanisms to create a cohesive and user-friendly smart pump.

4. Ensuring the system's reliability, accuracy, and safety for potential clinical use.

This project will systematically address these challenges through a multidisciplinary approach, combining principles from biomedical engineering, optics, data science, and endocrinology. The research will involve:

1. Design and optimization of an NIR sensor array for glucose detection in human tissue.
2. Development and validation of algorithms for processing NIR spectral data and predicting glucose levels.
3. Integration of the NIR sensing system with a prototype insulin pump.
4. In vitro and preliminary in vivo testing to assess system performance and accuracy.

The ultimate goal of this research is to create a more user-friendly and effective diabetes management tool that could significantly improve quality of life for individuals with diabetes. By combining non-invasive glucose monitoring with automated insulin delivery, this smart pump system has the potential to reduce the burden of diabetes management, improve glycemic control, and potentially decrease long-term complications associated with the disease.

1.2 STATEMENT OF THE PROBLEM

With the alarming increase in diabetic cases globally, there is an urgent need for an accurate and reliable continuous glucose monitoring (CGM) system. Current CGM technologies, while beneficial, often face challenges such as sensor accuracy, user discomfort, and data interpretation complexities. Additionally, there is a critical need for an insulin delivery system that not only administers insulin effectively but also ensures that cells can adequately utilize the insulin provided.

This research aims to address these pressing issues by developing a **Smart Insulin Pump** that integrates near-infrared (NIR) sensors. The proposed solution seeks to enhance the accuracy of glucose monitoring and improve the efficiency of insulin delivery. By leveraging NIR sensors, the smart insulin pump will provide real-time, non-invasive glucose measurements, thereby reducing the physical and psychological burden on patients. Furthermore, the integration of advanced algorithms will facilitate precise insulin dosing, optimizing metabolic control and improving the overall quality of life for individuals with diabetes.

1.3 AIM AND OBJECTIVES OF THE STUDY

The aim of this study is to develop a cost-effective, smart insulin pump for automated blood glucose control and management.

In order to achieve this aim, we plan

- To Incorporate real-time continuous glucose monitor
- To Develop control algorithms to automate insulin delivery
- To develop a mobile application for remote monitoring/alerts.
- To Test and validate the automated insulin pump system

1.4 SCOPE OF THE STUDY

This study focuses on the development of a smart insulin pump for diabetic patients using NIR sensors. The scope includes:

1. Design and integration of NIR sensors for non-invasive glucose monitoring.
2. Development of a closed-loop control system for automated insulin delivery.
3. Creation of algorithms for glucose level estimation and insulin dosing.
4. Design of a user-friendly interface for data display and device control.
5. Implementation of wireless connectivity for data sharing.
6. Prototype development and assembly.
7. Initial performance evaluation through laboratory testing.

The study will be limited to the technical development and preliminary testing of the device. Clinical trials and long-term patient studies are beyond the scope of this undergraduate project.

1.5 SIGNIFICANCE OF THE STUDY

This study on a smart insulin pump using NIR sensors is significant because:

1. It aims to improve patient comfort by offering non-invasive glucose monitoring.
2. It could enhance glucose control through continuous monitoring and automated insulin delivery.
3. It contributes to advancing smart medical device technology.
4. It may reduce the burden on patients in managing their condition.

5. It allows for real-time health monitoring and data sharing with healthcare providers.
6. It has the potential to lower long-term healthcare costs associated with diabetes management.
7. It promotes interdisciplinary research in biomedical engineering and related fields.
8. It provides a foundation for future studies in diabetes care technology.

1.6 DEFINITION OF TERMS

- Diabetes Mellitus: A metabolic disorder characterized by high blood glucose levels due to inadequate insulin production or ineffective insulin use.
- Insulin: A hormone that regulates blood glucose levels by facilitating the absorption of glucose into cells.
- Insulin Pump: A medical device that delivers insulin continuously to manage diabetes.
- Smart Insulin Pump: An advanced insulin pump that uses technology to automate and optimize insulin delivery.
- NIR Sensors: Near-Infrared sensors that use light to non-invasively measure glucose levels in the body.
- Glucose Monitoring: The process of measuring blood glucose levels.
- Closed-loop System: An automated system that adjusts insulin delivery based on continuous glucose monitoring without manual intervention.
- Glycemic Control: The management of blood glucose levels within a target range.
- Non-invasive: A method that doesn't require breaking the skin or entering the body.
- Real-time Data: Information that is delivered immediately after collection.

- Algorithms: A set of rules or calculations used by the device to make decisions about insulin delivery.
- Wireless Connectivity: The ability of the device to transmit data wirelessly to other devices or systems.

CHAPTER TWO

LITERATURE REVIEW

2.1 DIABETES

Diabetes mellitus is a group of metabolic disorders characterized by elevated blood glucose levels, or hyperglycemia, resulting from the body's inability to produce or effectively utilize insulin, a hormone responsible for regulating blood sugar (Peter et al., 2018)]. Hyperglycemia serves as the primary biomarker for the diagnosis of diabetes (Banday et al., 2020). Diabetes mellitus is associated with a broad range of clinical presentations, from being asymptomatic to ketoacidosis or coma, depending on the degree of metabolic disorder (Seino et al., 2010).

There are two main types of diabetes: type 1 and type 2. Type 1 diabetes is characterized by the body's inability to produce insulin, typically due to an autoimmune reaction that destroys the insulin-producing beta cells in the pancreas (Banday et al., 2020). Type 2 diabetes, on the other hand, is associated with a lack of serum insulin or poor uptake of glucose into the cells, leading to relative insulin deficiency (Nair, 2007).

The etiology of diabetes is multifactorial, involving both genetic and environmental factors (Seino et al., 2010). Type 2 diabetes mellitus is characterized by relative insulin deficiency caused by pancreatic cell dysfunction and insulin resistance (Widiasari et al., 2021).

2.2 INSULIN

Insulin therapy remains a cornerstone in the treatment of diabetes mellitus, with continuous research dedicated to enhancing both delivery methods and formulations (Zahra Baghban Taraghdari et al., 2019). The most effective management strategies currently include multiple daily insulin injections and continuous subcutaneous insulin infusion (CSII) (Muhammad Sarfraz Nawaz et al., 2017).

Various insulin analogues have been developed to better mimic the body's natural insulin response. These include rapid-acting, short-acting, long-acting, and intermediate-acting formulations (Chythanya C. Kutty et al., 2020). Insulin is essential not only for regulating blood glucose levels but also for modulating numerous physiological processes, making its proper synthesis and regulation vital for overall health (M. S. Rahman et al., 2021).

Recent advancements in insulin therapy have explored novel delivery methods such as oral and inhalant insulin. These innovations hold the potential to revolutionize diabetes management by offering more convenient and less invasive options for patients (Muhammad Sarfraz Nawaz et al., 2017).

Ongoing research is focused on developing insulin formulations and delivery systems that can more closely replicate the natural insulin response. The goal is to improve patient outcomes by enhancing the efficacy and convenience of insulin therapy.

2.3 GLUCOMETER

Recent years have witnessed significant advancements in diabetes management technology, substantially improving patient care. Colin and Paris (2012) report that blood glucose meters incorporating automated bolus calculators can enhance decision-making and potentially improve metabolic control for insulin-treated patients, provided adequate education is maintained. The integration of continuous glucose monitoring (CGM) and insulin pumps has led to the development of "smart" insulin pumps, which can automate insulin delivery to achieve more stable glycemic control while minimizing hypoglycemia (Grunberger, 2019).

Ramchandani and Heptulla (2012) highlight that these technologies, along with traditional glucose meters and CGM devices, represent significant progress in diabetes management. However, Grunberger (2019) notes that incorporating these technologies into daily clinical practice presents challenges, including appropriate patient selection and adaptation.

The literature emphasizes ongoing research and clinical trials exploring integrated systems to further improve patient outcomes and quality of life (Grunberger, 2019; Ramchandani & Heptulla, 2012). These studies suggest that while current technologies offer substantial benefits, future innovations may provide even more comprehensive solutions for diabetes management.

Several authors underscore the importance of patient education and support in maximizing the effectiveness of these technologies (Colin & Paris, 2012; Grunberger, 2019). The literature also points to the need for healthcare providers to carefully consider individual patient needs and capabilities when recommending advanced diabetes management tools.

As the field of diabetes technology continues to evolve, researchers are investigating more sophisticated integrated systems, including closed-loop or artificial pancreas systems, which could potentially automate glucose monitoring and insulin delivery with minimal user intervention (Grunberger, 2019; Ramchandani & Heptulla, 2012).

This body of literature highlights the potential of technological advancements to significantly improve diabetes care, while also acknowledging the challenges in implementation and the ongoing need for further research and development in this rapidly evolving field.

2.4 CONTINUOUS GLUCOSE MONITORING

Continuous glucose monitoring (CGM) has emerged as a transformative technology in the field of diabetes management. Unlike traditional blood glucose monitoring methods that provide only intermittent snapshots of a patient's glucose levels, CGM offers a continuous stream of real-time data. This comprehensive approach has significantly enhanced the ability to detect and respond to both hypoglycemic (low blood sugar) and hyperglycemic (high blood sugar) episodes, which are critical concerns in diabetes care (Cappon et al., 2019).

The evolution of CGM technology has led to the development of various types of devices to meet different patient needs. Professional CGM devices are typically used in clinical settings for short-term, retrospective analysis of a patient's glucose patterns. Personal CGM systems, on the other hand, provide real-time data directly to patients, allowing for immediate adjustments in diabetes management. Additionally, the integration of CGM sensors with insulin pumps has created more seamless and automated diabetes management systems, representing a significant advancement in care (Klonoff et al., 2017).

The benefits of CGM use have been well-documented in numerous studies. Research consistently shows that patients with diabetes who use CGM experience improvements in several key areas of diabetes management. These include lower HbA1c levels (a measure of long-term blood sugar control), reduced frequency and severity of hypoglycemic events, and an overall enhancement in quality of life. These positive outcomes underscore the transformative potential of CGM technology in diabetes care (Klonoff et al., 2017).

Beyond these measurable clinical outcomes, CGM systems offer valuable psychological and behavioral benefits. Patients often report reduced fear of hypoglycemia, as the continuous monitoring and alert features provide an added layer of security. Furthermore, the real-time feedback from CGM devices has been shown to improve patients' self-management skills, empowering them to make more informed decisions about their diet, exercise, and medication regimens (Chaithanya, 2023).

However, it's important to note that the implementation of CGM technology is not without challenges. The vast amount of data generated by these devices can be overwhelming for both patients and healthcare providers to manage and interpret effectively. This highlights the need for improved data management tools and strategies to maximize the benefits of CGM technology (Chaithanya, 2023).

The development of CGM has also spurred innovation in related areas of diabetes management. For instance, the continuous stream of glucose data has enabled the creation of predictive alerts that can warn patients of impending hypo- or hyperglycemic events before they occur. Additionally, when integrated with insulin pumps, CGM data can be used to automatically adjust insulin delivery, a feature known as automatic insulin attenuation. These advancements represent significant steps towards more proactive and personalized diabetes management (Facchinetti, 2016).

Looking to the future, ongoing research in CGM technology focuses on several key areas. Efforts are being made to further improve the accuracy of glucose measurements, addressing one of the primary concerns with current CGM systems. Researchers are also working to address patient-reported challenges, such as improving the comfort and longevity of CGM sensors. Perhaps most excitingly, the exploration of novel sensing technologies is underway. This includes research into optical signaling methods and advanced biosensors, which could potentially offer even more precise and less invasive glucose monitoring options in the future (Chaithanya, 2023).

In conclusion, continuous glucose monitoring has significantly transformed the landscape of diabetes management. By providing real-time, comprehensive glucose data, CGM technology has enabled more precise and personalized care for patients with diabetes. While challenges remain, ongoing research and technological advancements continue to push the boundaries of what's possible in glucose monitoring and diabetes management, promising even greater improvements in patient care in the years to come.

2.4 INSULIN PUMP

Insulin pump therapy, also known as continuous subcutaneous insulin infusion (CSII), has gained recognition as a highly effective approach for diabetes management. This method offers significant benefits in terms of glycemic control and patient flexibility (Sora et al., 2019; Nimri et al., 2019). Insulin pumps deliver insulin continuously and allow for bolus doses, making them suitable for individuals with type 1 diabetes and some with type 2 diabetes (Sora et al., 2019).

Technological advancements have played a crucial role in enhancing insulin pump therapy. The development of smart insulin pens equipped with connectivity features represents a significant leap forward. These innovations can improve diabetes self-management and boost patient confidence (Heinemann et al., 2021). Recent research indicates that insulin pump therapy can reduce the incidence of hypoglycemia and may also lower the risk of both microvascular and macrovascular complications (Nimri et al., 2019).

The integration of continuous glucose monitoring (CGM) systems with insulin pumps has further optimized diabetes treatment. The ability to download and analyze data from these devices allows for more precise adjustments to therapy, leading to better overall outcomes (Nimri et al., 2019; Satuluri et al., 2023). As technology continues to advance, future developments are likely to include closed-loop systems, also known as artificial pancreas systems, and decision support tools. These innovations hold the potential to further enhance metabolic control and improve the quality of life for individuals with diabetes (Nimri et al., 2019).

In summary, insulin pump therapy, supported by continuous technological advancements, offers a promising avenue for improving diabetes management. The ongoing evolution of these technologies is expected to bring about even greater improvements in patient outcomes and overall quality of life.

2.5 INSULIN PATCH

Recent advancements in insulin delivery systems for diabetes management have led to the development of innovative technologies such as patch pumps and artificial pancreas systems. Patch pumps, exemplified by the OmniPod®, provide a tubeless, lightweight, and discrete method of insulin delivery. These features significantly enhance glycemic control and improve the quality of life for individuals with type 1 diabetes (Filipczak et al., 2022).

These patch pumps are anticipated to be integral components of future artificial pancreas systems. The goal of these systems is to automate glucose regulation by integrating continuous glucose monitors, insulin pumps, and sophisticated control algorithms (Priya & Kalyani, 2019).

Innovative approaches in this field include the development of glucose-responsive pectin insulin patches. These patches have demonstrated potential in preclinical studies for mitigating hyperglycemia and reducing diabetes-related complications (Krisnamurti et al., 2022).

Moreover, wearable patch delivery systems are evolving to incorporate sensing capabilities and multifunctional electronics on flexible platforms. This evolution holds promise for personalized diagnostic and therapeutic applications in diabetes management (Ahmad et al., 2021).

These advancements mark significant progress towards more effective and user-friendly diabetes treatment options, paving the way for improved patient outcomes and enhanced quality of life.

2.6 IDENTIFICATION OF GAPS IN THE CURRENT KNOWLEDGE

Recent advancements in diabetes management have led to the development of smart insulin pumps integrated with continuous glucose monitoring (CGM) systems, offering potential improvements in glycemic control and reduced hypoglycemia risk (Grunberger, 2019). However, research has identified gaps in healthcare providers' knowledge and understanding of these technologies, particularly among pediatric endocrinology trainees (Marks et al., 2019). This lack of expertise may contribute to suboptimal patient outcomes despite increased technology adoption. To address these issues, high-priority research needs have been identified, including studies on the effectiveness of various insulin delivery and glucose monitoring methods in different patient populations (Yeh et al., 2013). Additionally, challenges in incorporating diabetes technology into clinical practice and patient selection criteria require further investigation (Grunberger, 2019). Developing formal educational curricula for healthcare providers and conducting targeted research may improve patient care and outcomes in the evolving landscape of diabetes management.

Continuous glucose monitoring (CGM) has become an important tool for managing diabetes, particularly for insulin-requiring patients (Klonoff et al., 2017). CGM devices provide real-time glucose data, historical trends, and alerts for extreme glucose levels, enabling better self-management and improved outcomes, especially for type 1 diabetes patients (Klonoff et al., 2017). However, challenges persist in CGM adoption and effectiveness. These include physical limitations, stigma, alarm fatigue, and the complexity of interpreting large volumes of data (Barnard-Kelly et al., 2024). Adolescents using insulin pumps face particular challenges in self-management, with key behaviours like blood glucose monitoring and bolus frequency influencing glycemic outcomes (Faulds et al., 2021). To address these issues, researchers are working on developing more user-friendly CGM systems and pump control algorithms, such as the SmartCGMS platform, which aims to unify glucose monitoring and control across various devices (Martin Ubl & T. Koutny, 2019).

CHAPTER THREE

METHODOLOGY

3.1 Overview

This Section details the development process chosen for the construction of a smart insulin pump for monitoring glucose levels, The device can be divided into three components, The First component is the Near Infrared Radiation (NIR) glucose measuring unit whose job is to measure the glucose concentration using NIR technology, this component contains the microcontroller, Infrared Led and Photodiode to measure glucose concentration and to make a further decision based on these data. The second section includes the mobile application, which serves as a real-time graphical interface for monitoring the information gathered by the NIR sensor, the mobile application maintains a two-way communication with the NIR Sensor via wireless Bluetooth communication. The third section which is the Insulin pump, administers insulin with the mechanism of converting a rotational motion to a translational motion, slowly pushing in a plunger and delivering insulin to a diabetic user

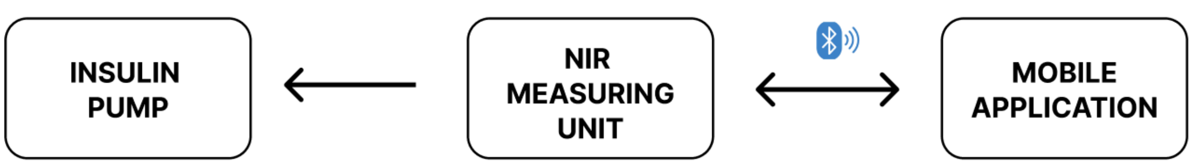

FIG 3.1 Block diagram of the component communication in the Smart insulin pump system

3. 2 NEAR-INFRARED RADIATION GLUCOSE MEASURING UNIT

3.2.1 NEAR-INFRARED RADIATION

Near-infrared (NIR) radiation is a type of electromagnetic radiation with wavelengths longer than visible light but shorter than mid-infrared light. It has wavelengths ranging from approximately 780 nanometers (nm) to 2500 nm

Near-infrared imaging is beneficial in non-invasive diagnostics and offers several advantages over traditional radiological techniques due to its non-ionizing nature (Seminars Only, 2023)

NIR is utilized in techniques such as near-infrared spectroscopy (NIRS), which measures oxygen levels in tissues and helps detect anomalies. This method is particularly valuable in assessing blood flow and oxygenation in various medical conditions, including those related to blood clots and vascular health

NIR spectroscopy works by analyzing how near-infrared light interacts with glucose molecules in the blood. Specific wavelengths of NIR light are absorbed by glucose, enabling for the estimation of blood glucose levels based on the absorption characteristics of the light.

NIR light can penetrate the skin to reach the dermis, where it interacts with blood components. This capability enables the measurement of glucose levels without the need for blood samples, making it a less invasive option for patients

Glucose molecules absorb specific wavelengths of near-infrared light, typically in the range of 1000-2500 nm (Uwadaira & Ikehata, 2018). The absorption is directly related to the glucose concentration in the blood and The relationship between the absorbed light intensity and glucose concentration is described by Beer-Lambert's law.

Beer-Lambert's law states that the absorbance of light is proportional to the concentration of the absorbing species. The fingertip has a lot of capillary density hence it is a good site for glucose measurement

$$A(\lambda) = Log_{10}\left(\frac{I_0(\lambda)}{I(\lambda)}\right)$$

$$A(\lambda) = \epsilon(\lambda) \cdot c \cdot l$$

$A(\lambda)$ = is the absorbance at wavelength λ.

$I_0(\lambda)$ = is the intensity of the incident light at wavelength λ.

$I(\lambda)$ = is the intensity of the transmitted light at wavelength λ.

$\epsilon(\lambda)$ = is the molar absorptivity (or extinction coefficient) at wavelength λ.

c = is the concentration of the glucose in the sample.

l = is the path length that the light travels through the sample.

Glucose exhibits distinct absorption properties in the near-infrared region. By analyzing the NIR spectrum at different wavelengths, the glucose concentration can be determined. The absorbance ($A(\lambda)$) represents the amount of light absorbed by glucose molecules in the sample, which correlates with the concentration of glucose.

Certain wavelengths in the NIR region (typically between 700 nm and 2500 nm) are chosen where glucose has distinct absorption features.

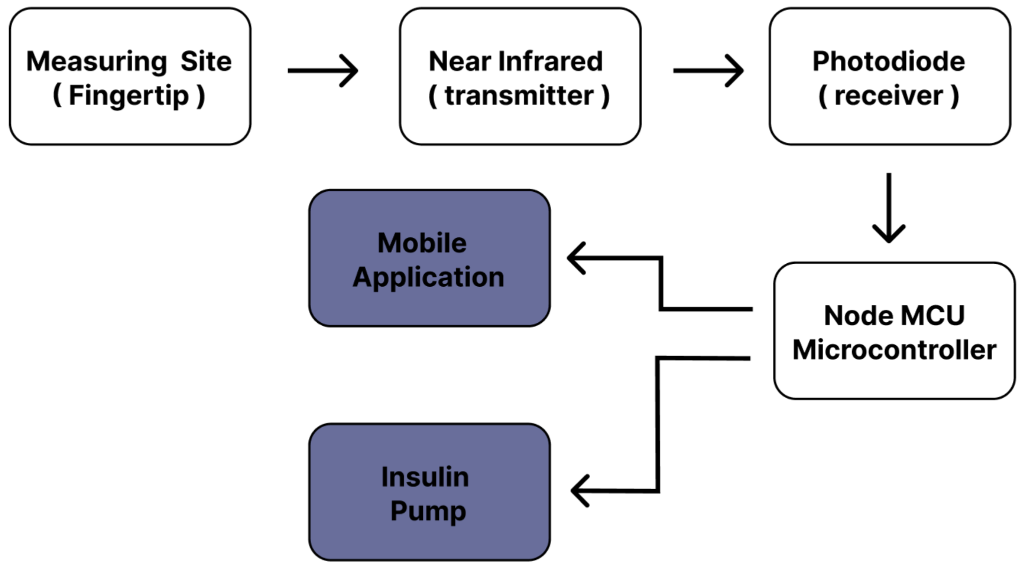

FIG 3.2 Block diagram of the Near Infrared Radiation (NIR) glucose measuring unit.

3.2.2 INFRARED LED

The Infrared Light Emission Diode used has a wavelength of 850nm, a forward current of 50mA, and a reverse current of 10 uA

The Anode (positive terminal) is the longer legs while the Cathode (negative terminal) is the shorter legs

Their operation is based on the principle of electroluminescence, similar to visible light LEDs, but they emit light in the infrared spectrum, which is invisible to the human eye.

When a voltage is applied across the semiconductor material, typically made of gallium arsenide (GaAs) or aluminum gallium arsenide (AlGaAs), electrons and holes recombine. This recombination releases energy in the form of infrared light. The specific wavelength of the emitted infrared light depends on the semiconductor materials used and their bandgap energy. IR

LEDs can emit light across different regions of the infrared spectrum, including near-infrared (NIR), mid-infrared (MIR), and far-infrared (FIR).

FIG 3.3 An Infrared LED(Light dependent diode)

3.2.3 PHOTODIODE

A photodiode is a semiconductor device that converts light into an electrical current. Its working principle is based on the photoelectric effect, where absorbed photons generate electron-hole pairs that produce a current flow when the photodiode is reverse-biased.

When photons of sufficient energy strike the photodiode, they create electron-hole pairs through the photoelectric effect. If the absorption occurs in the depletion region or within one diffusion length of it, the built-in electric field sweeps the carriers apart, generating a photocurrent.

The total current is the sum of the dark current (current in the absence of light) and the photocurrent

FIG 3.4 A Photodiode

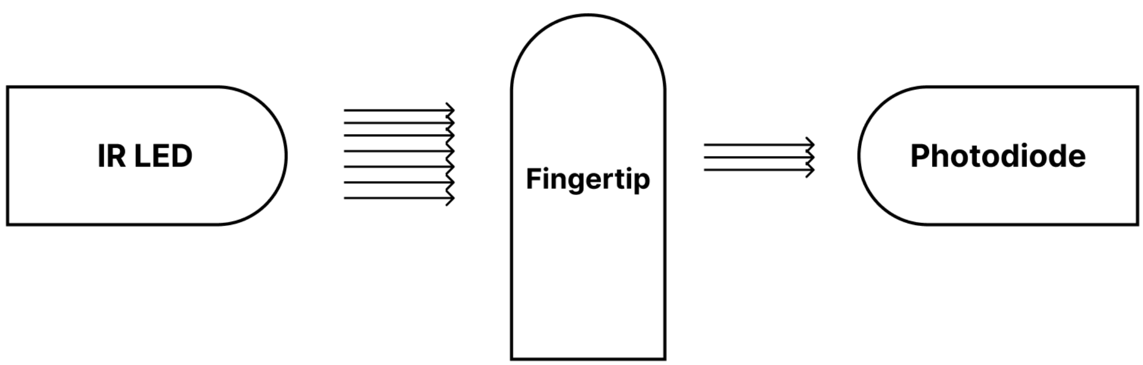

FIG 3.5 Block Diagram of the configuration between the IR LED and the Photodiode

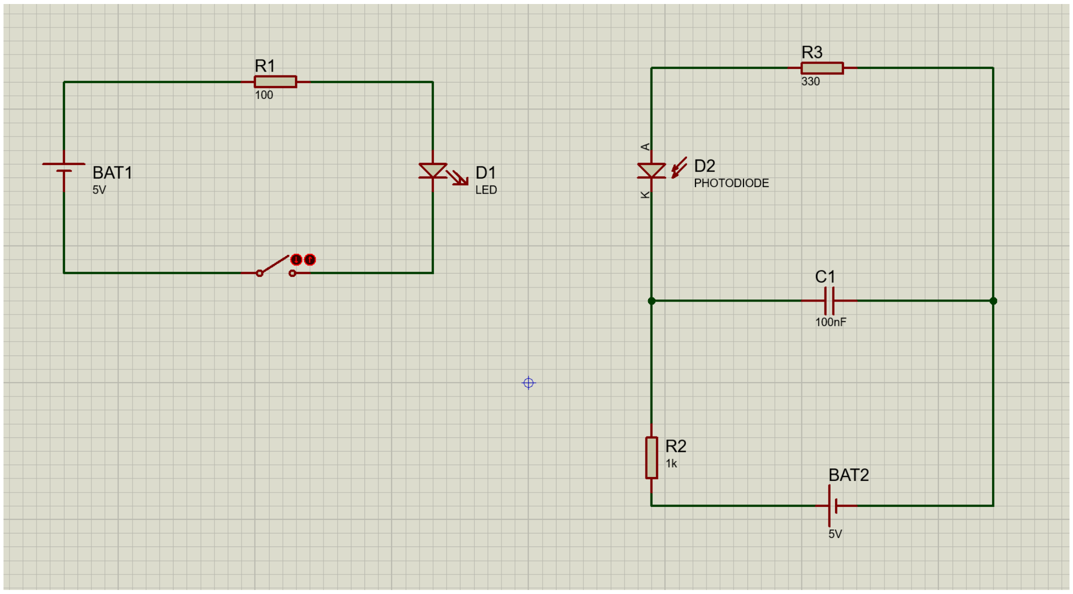

FIG 3.6 Schematic Diagram of the configuration between the IR LED and the Photodiode

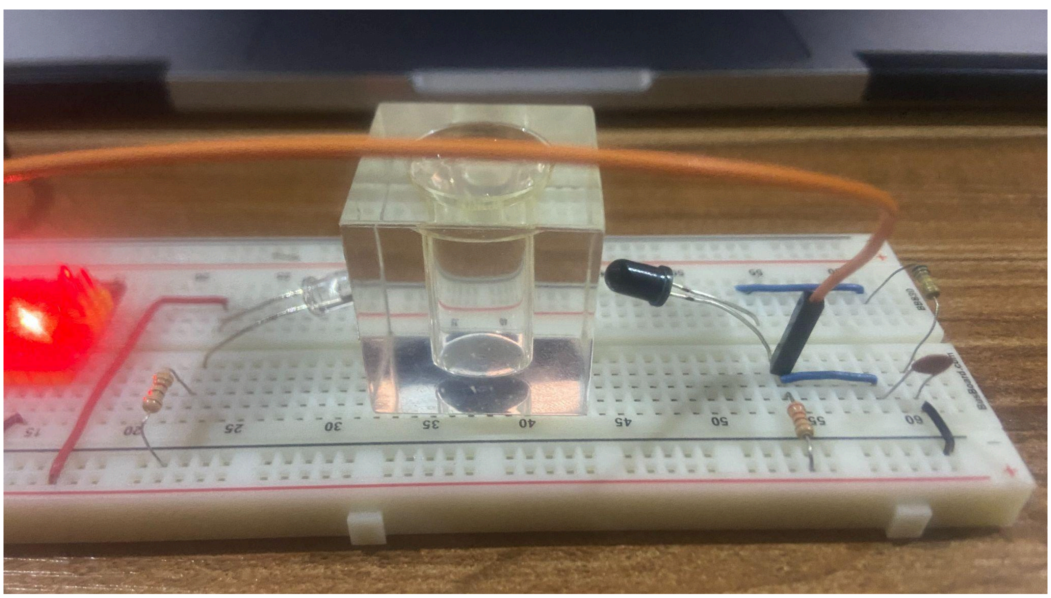

FIG 3.7 NIR setup with different concentration of glucose

3.2.4 THE ESP 32 MICROCONTROLLER

The Arduino ESP32 is a popular microcontroller board that combines Wi-Fi and Bluetooth connectivity with low-power consumption, making it ideal for Internet of Things (IoT) applications. It offers dual-core processing, a wide range of peripherals, and extensive support for various sensors and actuators (Maier et al., 2019). Recent research has demonstrated the ESP32's effectiveness in diverse fields, including environmental monitoring (Zhang et al., 2022), wearable health devices (Rodrigues et al., 2023), and smart home systems (Lee & Park, 2020). The Arduino ESP32's combination of features, performance, and ease of use continues to drive innovation in embedded systems and IoT development.

Table 3.1: ESP 32 Microcontroller specifications

Microcontroller	Tensilica 32-bit RISC CPU Xtensa LX106
Operating Voltage	3.3V
Input Voltage	7-12V
Analog Input Pins	1
Digital I/O Pins	16
Universal Asynchronous Receiver/Transmitter (UARTs)	1
Serial Peripheral Interfaces	1
Inter-Integrated Circuits	1
Flash Memory	4 MB
SRAM	64 KB
Clock Speed	80 MHz
Crystal Oscillator	40 Mhz

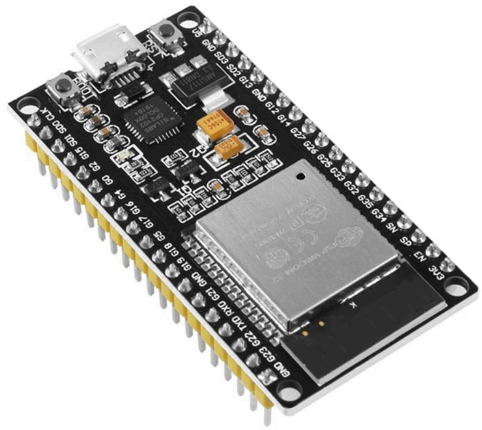

FIG 3.8 ESP32 Microcontroller

```
int irpin = A0;
int value = 0;
float voltval = 0;
void setup() {
  // put your setup code here, to run once:
  pinMode(A0, INPUT);
  Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
  value = analogRead(irpin);
  Serial.print("PhotoDiode reading: ");
  voltval = map(value, 0, 1023, 0, 5);
  Serial.println(voltval);
  delay(1000);
}

```

FIG 3.9 Arduino code for reading glucose val with an NIR sensor

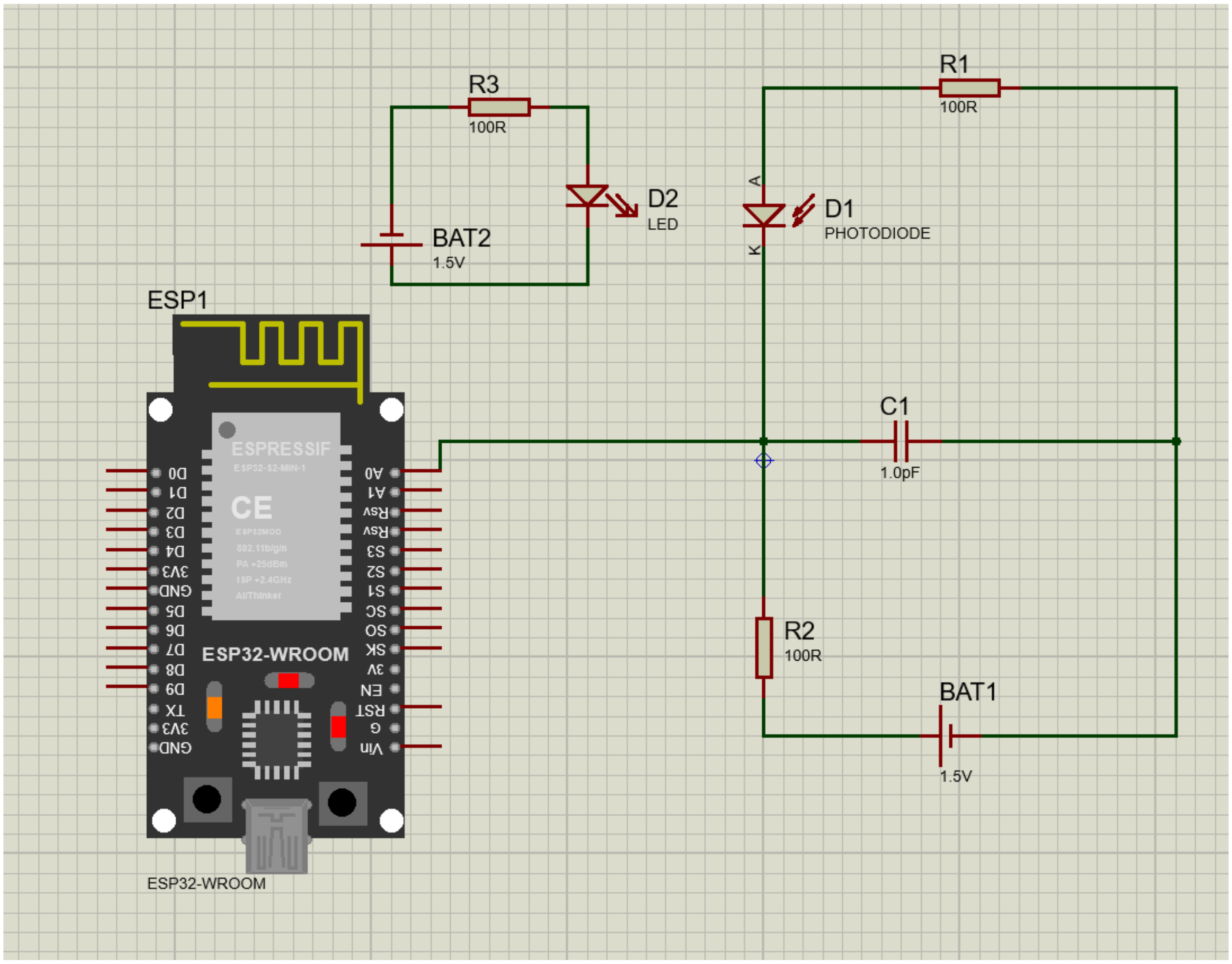

FIG 3.10 Schematic Diagram of the configuration between the NIR and the ESP32

3.3 SOFTWARE APPLICATION

The software application developed for this project is a mobile application created using React Native technology whose primary purpose is to interface and communicate with the NIR sensor component, facilitating the real-time monitoring of glucose concentration. The application is meticulous with two components: the logic layer and the user interface (UI) layer.

The logic layer of the application is responsible for a stable Bluetooth connection with the NIR sensor. It ensures that data is accurately received and processed, allowing the application to reliably consume the information provided by the sensor.

The UI layer is dedicated to the accurate presentation of the data within the application. The UI is designed to provide users with a clear and accessible view of the glucose concentration readings in real-time. This visual representation is essential for enabling users to monitor their glucose levels continuously, making the application an effective tool for diabetes management.

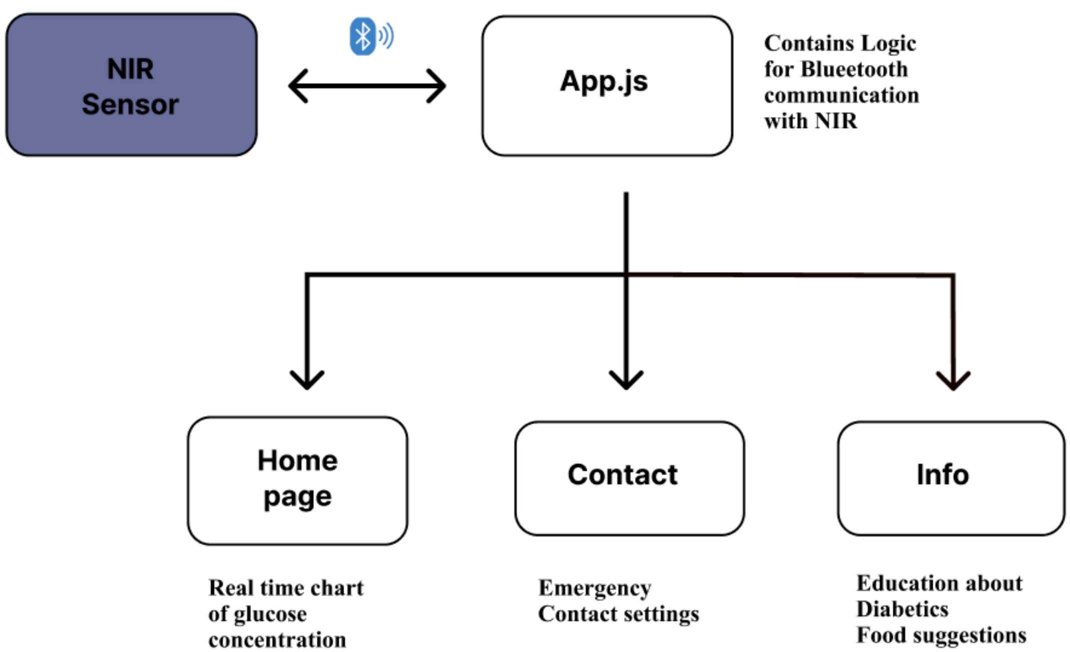

FIG 3.11 Block Diagram Software Architecture

3.3.1 REACT NATIVE

React Native is a popular open-source framework developed by Facebook for building cross-platform mobile applications using JavaScript and React allowing developers to create native mobile apps for both iOS and Android platforms using a single codebase.

There are two main workflows for developing React Native applications: Expo and React Native CLI.

Expo provides a managed development environment with a set of pre-built components and services, making it easier for beginners to get started quickly. Expo offers a mobile client app called Expo Go, that allows developers to run and test their applications on real devices without the need for complex setup or builds

The React Native CLI offers a more flexible and customizable approach, giving developers full control over the native code and the ability to add custom native modules. This workflow requires more setup but provides greater flexibility for advanced use cases and integration with native APIs.

In this project, the Expo workflow was used because of the ease of setup and speed in testing during development.

One benefit of utilizing native mobile development features is the ability to access core functionalities such as the camera, file storage, Wi-Fi, Bluetooth, and more. In this project, the Bluetooth functionality was employed to interact with the device's hardware component. Although React Native does not natively support Bluetooth, third-party libraries can be used to enable Bluetooth capabilities. A widely used library, and the one implemented in this project, is 'react-native-ble-plx.'

```
import { StatusBar } from "expo-status-bar";
import { useState, useEffect, useRef } from "react";
import { StyleSheet, View, PermissionsAndroid } from "react-native";
import { BleManager } from "react-native-ble-plx";
import { atob } from "react-native-quick-base64";

import Home from "./pages/Home";


const bleManager = new BleManager();

const SERVICE_UUID = "4fafc201-1fb5-459e-8fcc-c5c9c331914b";
const STEP_DATA_CHAR_UUID = "beefcafe-36e1-4688-b7f5-00000000000b";

export default function App() {
  const [deviceID, setDeviceID] = useState(null);
  const [connectionStatus, setConnectionStatus] = useState("Searching...");
  const [glucoselevel, setGlucoseLevel] = useState(0);

  const deviceRef = useRef(null);

  const searchAndConnectToDevice = () => {
    bleManager.startDeviceScan(null, null, (error, device) => {
      if (error) {
        console.error(error);
        setConnectionStatus("Error searching for devices");
        return;
      }
      if (device.name === "Smartinsulin") {
        bleManager.stopDeviceScan();
        setConnectionStatus("Connecting...");
        connectToDevice(device);
      }
    });
  };

  useEffect(() => {
    searchAndConnectToDevice();
  }, []);

  useEffect(() => {
    const subscription = bleManager.onDeviceDisconnected(
      deviceID,
      (error, device) => {
        if (error) {
          console.log("Disconnected with error:", error);
```

```
      }
      setConnectionStatus("Disconnected");
      console.log("Disconnected device");
      setStepCount(0); // Reset the step count
      if (deviceRef.current) {
        setConnectionStatus("Reconnecting...");
        connectToDevice(deviceRef.current)
          .then(() => setConnectionStatus("Connected"))
          .catch((error) => {
            console.log("Reconnection failed: ", error);
            setConnectionStatus("Reconnection failed");
          });
      }
    }
  );
  return () => subscription.remove();
}, [deviceID]);

const connectToDevice = (device) => {
  return device
    .connect()
    .then((device) => {
      setDeviceID(device.id);
      setConnectionStatus("Connected");
      deviceRef.current = device;
      return device.discoverAllServicesAndCharacteristics();
    })
    .then((device) => {
      return device.services();
    })
    .then((services) => {
      let service = services.find((service) => service.uuid === SERVICE_UUID);
      return service.characteristics();
    })
    .then((characteristics) => {
      let stepDataCharacteristic = characteristics.find(
        (char) => char.uuid === STEP_DATA_CHAR_UUID
      );
      setStepDataChar(stepDataCharacteristic);
      stepDataCharacteristic.monitor((error, char) => {
```

```
          if (error) {
            console.error(error);
            return;
          }
          const rawStepData = atob(char.value);
          console.log("Received step data:", rawStepData);
          setGlucoseLevel(rawStepData);
        });
      })
      .catch((error) => {
        console.log(error);
        setConnectionStatus("Error in Connection");
      });
  };
  return (
    <View style={styles.container}>
      <Home connectionStatus={connectionStatus} glucoselevel={glucoselevel} />
      <StatusBar style="auto" />
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: "#fff",
    padding: 20,
  },
});
```

FIG 3.12 React Native code of the software

The entry point of the application is the **App.js** file, which contains the **App** function. This function is the first to be executed when the application starts

Following this, the **searchAndConnectToDevice** is the next function to be executed, this function is triggered by the **useEffect** hook function provided by React. The useEffect function ensures that searchAndConnectToDevice runs as soon as the component is mounted. The searchAndConnectToDevice function leverages the bleManager package to scan for all nearby Bluetooth devices. Once the devices are identified, the function calls **connectToDevice**.

The **connectToDevice** checks if the devices are pairable with the unique service uuid, If a device is pairable, a connection is established, enabling two-way communication between the React Native mobile application and the hardware device.

Finally, the App component returns JSX (JavaScript XML), which manages the user interface (UI). The UI includes navigation to different pages, features a chart section that displays glucose levels over time and an information section that educates users on diabetics

3.3.2 ARDUINO

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It's widely used by hobbyists, educators, and professionals to build a variety of projects, from simple circuits to complex devices. An Arduino board would contain a microcontroller, input/output pins, power supply, and communication interfaces.

Arduino uses a simplified version of C++. The Arduino language adds some libraries and functions to standard C++ to make hardware interaction easier.

Arduino has various development environments like the Arduino IDE and Platform IO. In this project, the Platform IO environment was used. The advantages of using the Platform IO environment include easy integration with popular editors like Visual Studio Code, Allowing for multiple-platform support, and seamless integration with version controls like GIT.

```
#include <EEPROM.h>
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>
#define SERVICE_UUID "4fafc201-1fb5-459e-8fcc-c5c9c331914b"
#define GLUCOSE_DATA_CHAR_UUID "beefcafe-36e1-4688-b7f5-00000000000b"

BLECharacteristic *pGlucoseDataCharacteristic;

void setup() {
        // put your setup code here, to run once:
        Serial.begin(9600);
        delay(2000);
        // Create BLE device, server, and service
        BLEDevice::init("Glucose-Sense");
        BLEServer *pServer = BLEDevice::createServer();
        BLEService *pService = pServer->createService(SERVICE_UUID);

        // Create step data characteristic
        pGlucoseDataCharacteristic = pService->createCharacteristic(
            GLUCOSE_DATA_CHAR_UUID,
            BLECharacteristic::PROPERTY_READ | BLECharacteristic::PROPERTY_NOTIFY);
        pGlucoseDataCharacteristic->addDescriptor(new BLE2902());

        // Start BLE server and advertising
        pService->start();
        BLEAdvertising *pAdvertising = BLEDevice::getAdvertising();
        pAdvertising->addServiceUUID(SERVICE_UUID);
        pAdvertising->setScanResponse(true);
        pAdvertising->setMinPreferred(0x06);
        pAdvertising->setMinPreferred(0x12);
        BLEDevice::startAdvertising();
        Serial.println("BLE device is ready to be connected");

        EEPROM.begin(sizeof(int));
}

void loop() {
        String glucosedata = "connected to the react native application";
        pGlucoseDataCharacteristic->setValue(glucosedata.c_str());
        pGlucoseDataCharacteristic->notify();
        delay(2000);
}
```

FIG 3.13 Arduino code connecting the NIR glucose measuring Unit with the mobile application

The setup() function initializes the BLE device named "Glucose-Sense," creates a BLE server and service with specified UUIDs, and sets up a characteristic to hold step data that can be read and notified to connected clients. It also adds a descriptor to the characteristic for client configuration, starts the BLE service and advertising so that the device becomes discoverable, Finally, it indicates that the BLE device is ready for connections by printing a message to the serial monitor.

The loop() function continuously sends the glucose data to the React native mobile application every 2 seconds

3.4 INSULIN PUMP

Pumps are mechanical devices designed to move fluids by physical or mechanical action. The basic principle of pump mechanics involves creating a low-pressure area to draw in fluid and then increasing the pressure to expel it. There are various types of pumps, including centrifugal, positive displacement, and peristaltic pumps, each with specific mechanics suited to different applications (Tao et al., 2022)

An insulin pump is a medical device designed to deliver insulin continuously to manage diabetes. It mimics the function of a healthy pancreas by providing a steady basal rate of insulin and allowing for bolus doses as needed.

The mechanism of an insulin pump typically involves a reservoir for insulin, a pump motor, and a delivery system. Modern insulin pumps use a precision motor to push a plunger in the insulin reservoir, delivering small, accurate doses of insulin through tubing to a cannula inserted under the skin (Pickup, 2022).

3.4.1 STEPPER MOTOR

A stepper motor is an electromechanical device that converts electrical pulses into precise mechanical movements. Unlike traditional motors that rotate continuously, stepper motors move in discrete steps, allowing for controlled rotation and precise positioning. This makes them ideal for applications requiring accurate control of angular or linear position, such as in 3D printers, CNC machines, robotics, and automated camera platforms.

The working principle of a stepper motor is based on electromagnetism. The stator of a stepper motor is composed of multiple windings or coils arranged in phases. These windings are energized in a specific sequence to create a magnetic field. The rotor can be either a permanent magnet or a toothed iron piece. In hybrid stepper motors, the rotor typically has both magnetic and toothed elements.

When a current is passed through the stator windings, an electromagnetic field is generated. The rotor aligns itself with this magnetic field. By energizing the stator windings in a specific sequence, the magnetic field rotates, and the rotor follows this rotation step by step. Each time the stator's magnetic field moves to the next position, the rotor advances by one "step."

The direction, speed, and angle of rotation are controlled by the sequence and timing of the pulses sent to the stator windings. For example, increasing the pulse frequency increases the motor's speed.

The Stepper motor used in this project is the 28BYJ-48 and the driver used is the ULN2003

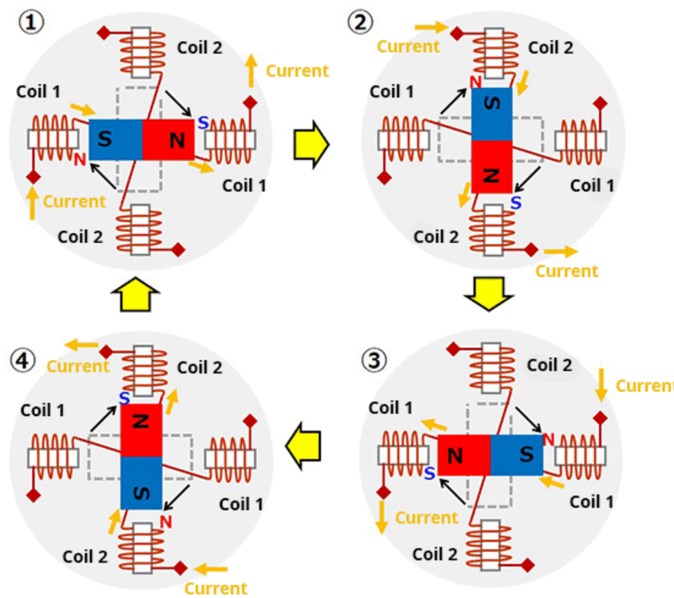

Fig 3.14 Internal workings of a stepper motor

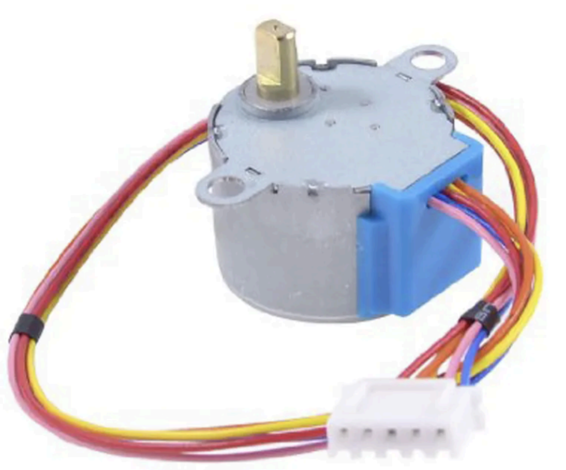

Fig 3.15 Stepper motor

3.4.2 STEPPER MOTOR DRIVER (ULN2003)

The ULN2003 stepper motor driver is a crucial component in controlling stepper motors, particularly in applications requiring precise motion control. This driver is an integrated circuit that includes an array of Darlington transistors, which are used to amplify and switch the current necessary to drive the motor's coils. The ULN2003 is designed to interface with low-power control signals from a microcontroller or other digital circuits, making it an ideal choice for driving stepper motors in a variety of projects, including robotics, CNC machines, and automated systems.

One of the significant advantages of the ULN2003 is its ability to manage the high current requirements of stepper motors while being controlled by the lower current outputs of a microcontroller. This capability ensures that the motor receives the necessary power to operate efficiently without overwhelming the control circuit. The driver includes built-in diodes for protection against back EMF (electromotive force), which can occur when the motor windings are de-energized, thereby safeguarding the microcontroller and other connected components from potential damage.

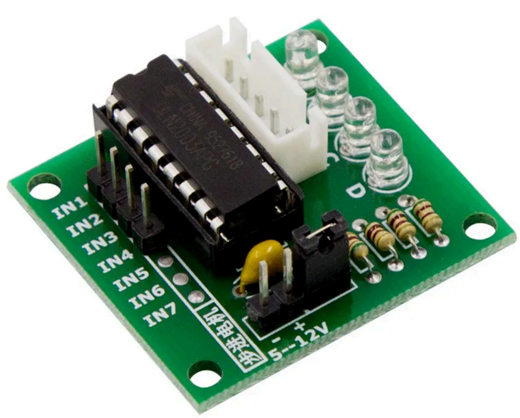

Fig 3.16 ULN2003 Stepper motordriver

3.4.3 GEARS

Gears are mechanical components with teeth that mesh together to transmit torque and rotational motion between shafts. They are fundamental in machines where controlled motion and power transmission are essential. Gears can change the direction of movement, increase torque, or adjust the speed of rotation in various mechanical systems, from watches and bicycles to industrial machines and vehicles.

When Two gears are in contact the driver gear can impact the torque and the speed of the driven gear. This influence is due to their Gear ratio. The Gear ratio is defined as the ratio of the number of teeth on two meshing gears.

$$Gear\ ratio\ = \frac{Number\ of\ teeth\ of\ the\ driven\ gear\ NB}{Number\ of\ teeth\ of\ the\ driver\ gear\ NA}$$

Hence

$$TorqueB\ = \ TorqueA\ *\ Gear\,ratio$$

$$SpeedB\ = \ \frac{SpeedA}{Gear\,ratio}$$

Where A is the Driver Gear and B is the Driven Gear

Fig 3.17 Driver gear and driven gear

In this project, rotational motion needs to be converted to linear motion to drive the pump. There are various ways to convert rotational motion to linear motion but the approach taken in this project is called the Rack and Pinion system.

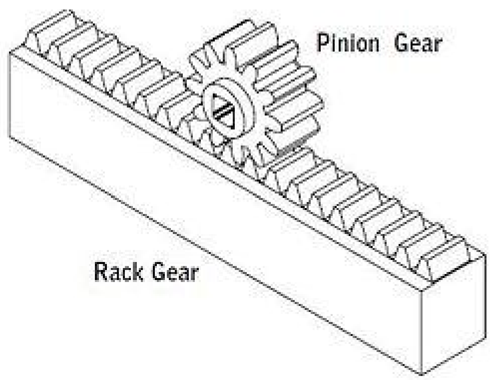

Fig 3.18 Pinion Gear and Rack Gear

The linear distance moved by the rack can be determined by the circumference of the pinion gear and the number of rotations. The formula is:

$Linear\ Distance\ =\ Number\ of\ rotation\ *\ Circumference\ of\ the\ pinion$

The volume of liquid administered is proportional to the length of the rack moved during the motor rotation.

Since the volume of a cylinder is $volume\ =\ \Pi r^2 h$

Then the volume administered is $volume\ =\ \Pi r^2 l$

Since $Linear\ Distance\ =\ Number\ of\ rotation\ *\ Circumference\ of\ the\ pinion$

Then the volume administered is

$$volume = \Pi r^2 * (Number\ of\ rotation * Circumference\ of\ the\ pinion)$$

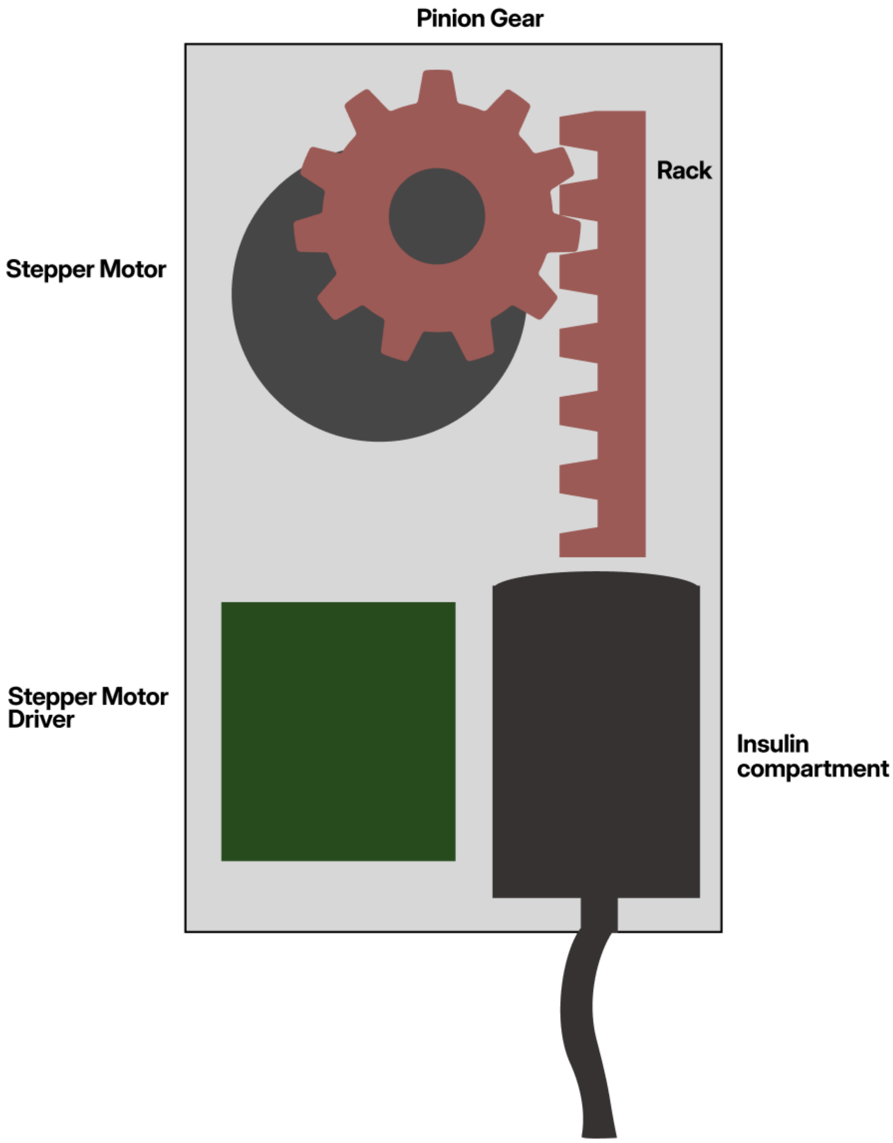

Fig 3.19 Block Diagram of the components of an Insulin pump

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 OVERVIEW

This section of the report provides details of the outcome of each component of the device, the NIR Glucose measuring unit, the software application and the insulin pump.

4.2 NEAR-INFRARED RADIATION GLUCOSE MEASURING UNIT

The NIR monitor was set up to measure the photodiode voltage at varying concentrations of glucose, and the following data was recorded

Table 4.1: Relationship between glucose concentration and photodiode voltage

Glucose Concentration (mg/dL)	**Photodiode Voltage (mV)**
70	1.524
90	1.793
110	2.132
130	2.476
150	2.688

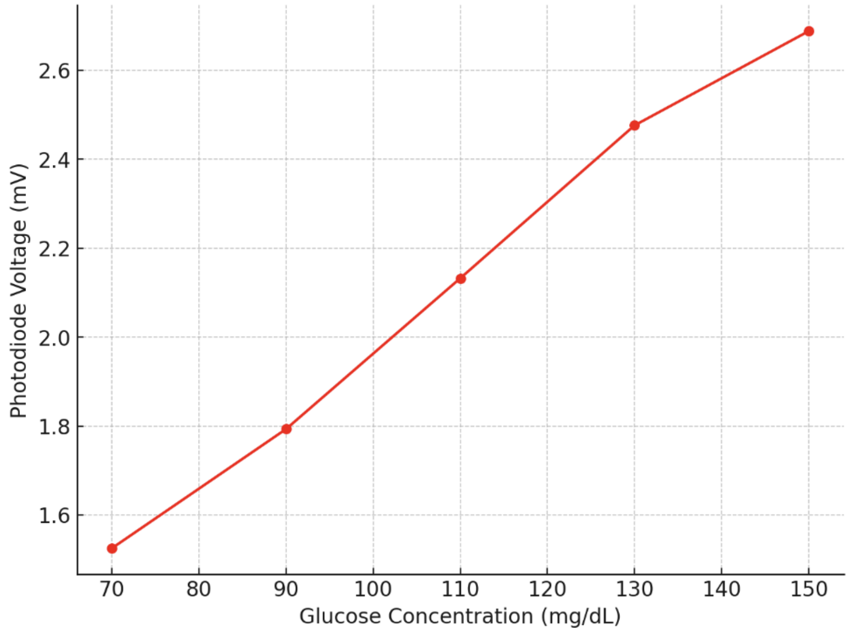

Fig 4.1 Graph showing the relationship between voltage and glucose concentration

The results presented in the table and graph indicate a generally linear relationship between glucose concentration and the photodiode voltage output. As the glucose concentration increases, the voltage output from the photodiode also increases. This trend aligns with the underlying principle that higher glucose concentrations result in greater absorption of near-infrared (NIR) light, reducing the amount of light reaching the photodiode and thus altering its voltage output.

Challenges Encountered

- Noise due to the electronics
- Photodiodes can also pick up light from other sources that are not the intended source.

4.3 SOFTWARE APPLICATION

The mobile application successfully displays real-time glucose levels measured by the NIR monitor. The chart feature allows users to track their glucose trends throughout the day, providing immediate feedback and helping them make informed decisions about their diet, activity, and insulin management.

The real-time chart is intuitive and provides a clear visual representation of glucose levels. This feature empowers users by giving them continuous access to their glucose data, allowing for better self-management and timely intervention when glucose levels start to rise.

The App contains an educational section that offers users articles on diabetes management, healthy eating, and lifestyle tips. This section is a valuable resource for users, helping them stay informed and make better health decisions.

Feedback indicates that users find the educational content relevant and helpful. The inclusion of up-to-date articles helps users stay informed about the latest developments in diabetes care.

The education section not only provides knowledge but also encourages healthier habits. Users report feeling more confident in managing their condition after engaging with the educational material.

The App also contains an Emergency Contact Feature that allows users to quickly reach out for help in case of a critical glucose event. This feature is designed to provide peace of mind to both users and their caregivers

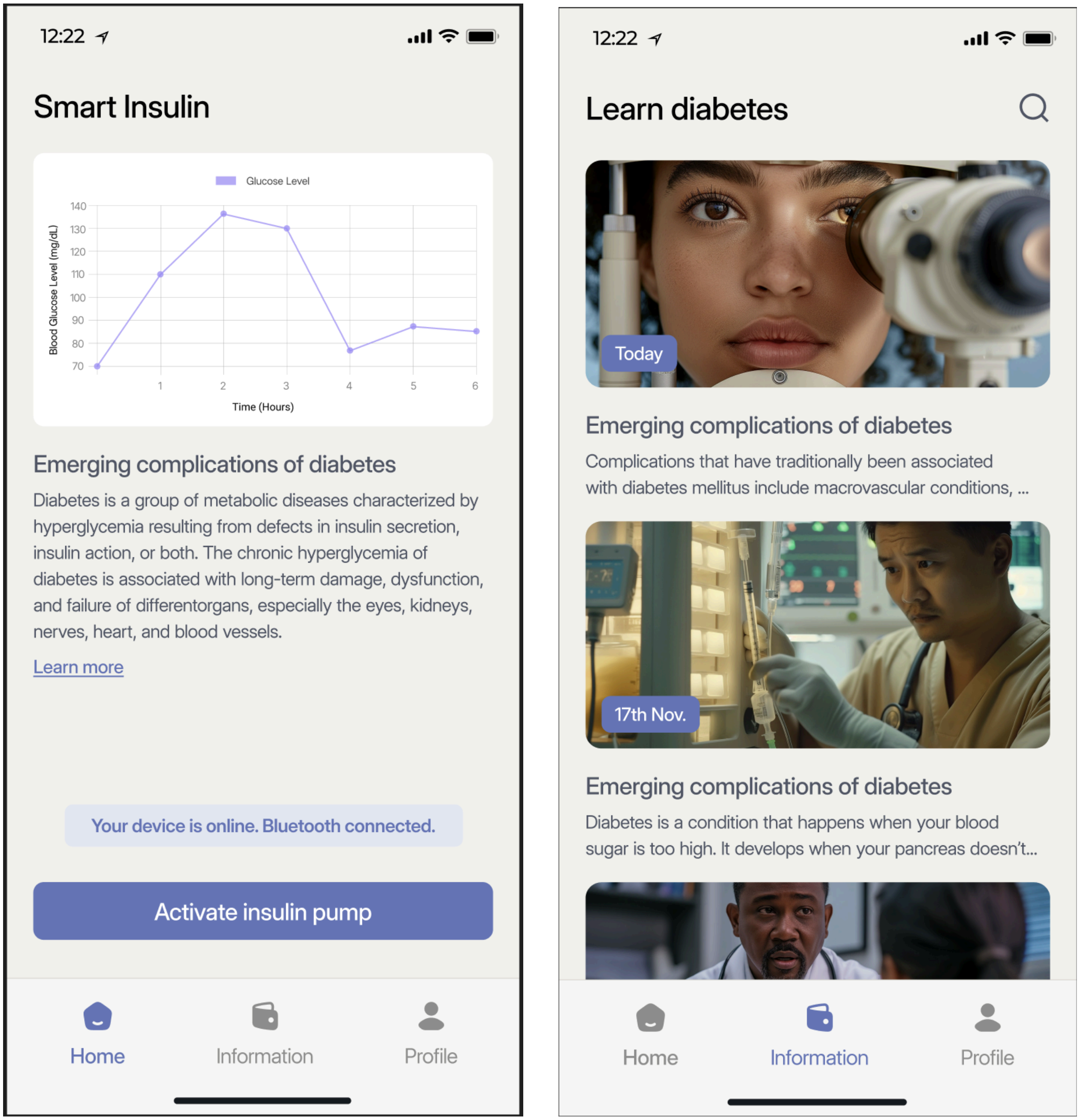

Fig 4.2 Mobile Application

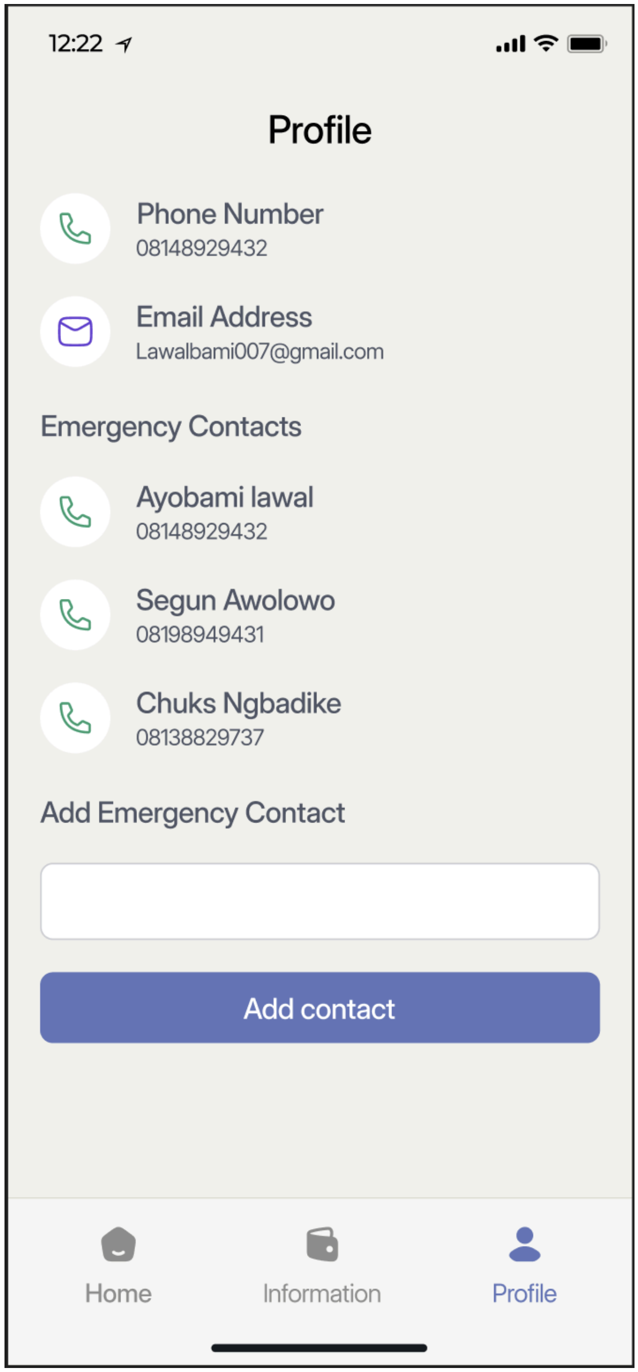

Fig 4.2 Mobile Application

4.4 INSULIN PUMP

The 3D design and simulation were created using the CAD software Fusion 360. The insulin pump utilizes a pinion gear and rack mechanism to convert rotational motion into translational motion. A stepper motor is employed to generate the rotational movement, as it provides sufficient torque to drive the gears. This causes the rack to move linearly, which, in turn, enables the plunger to deliver insulin.

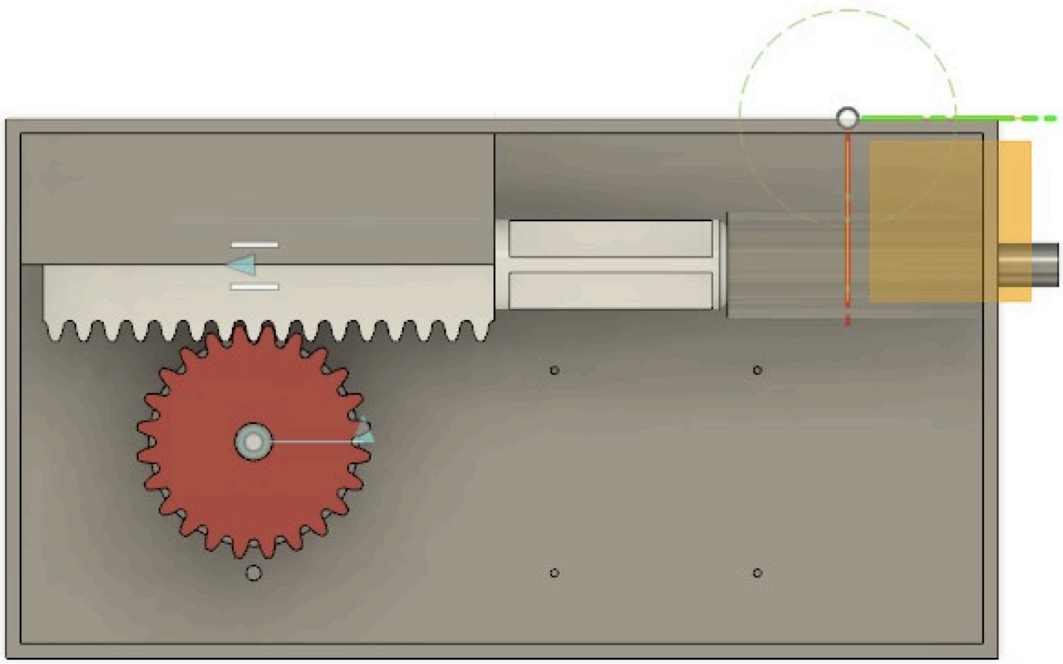

Fig 4.3 Top View of the 3D design

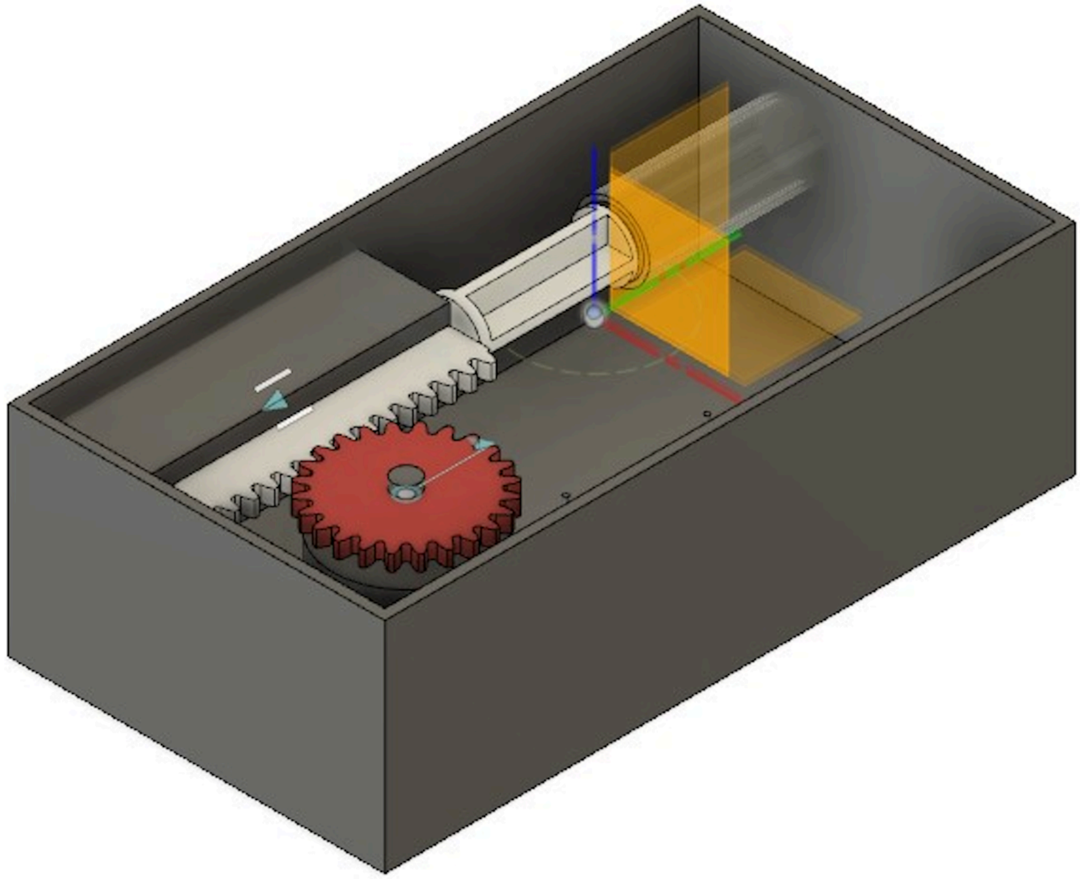

Fig 4.4 3D Design of the Device

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 CONCLUSION

This project successfully developed a smart insulin pump system for diabetic patients, integrating NIR sensors, a mobile application, and an automated insulin delivery mechanism. The system consists of three vital independent components:

1. A wearable sensing unit (wristwatch and a finger clip) utilizing NIR technology for non-invasive glucose monitoring.
2. A mobile application that displays real-time glucose readings and provides additional features for diabetes management.
3. An intelligent insulin pump that administers insulin based on the patient's specific requirements.

The integration of these components resulted in a comprehensive, user-friendly system that offers continuous glucose monitoring, data visualization, and automated insulin delivery. This approach demonstrates significant potential for improving glycemic control and enhancing the quality of life for diabetic patients by reducing the burden of manual monitoring and insulin administration.

The non-invasive nature of the NIR sensors addresses the discomfort associated with traditional glucose monitoring methods, while the mobile application provides patients with easy access to their health data. The automated insulin pump, working in tandem with the sensing unit and mobile app, offers a more precise and responsive method of insulin delivery.

These three components are designed to function seamlessly together, yet they are also easily interchangeable with similar devices. For instance, the NIR glucose monitoring module can be swapped out for a more accurate system without disrupting the functionality of the mobile application or the insulin pump. Similarly, any of the other components can be replaced with equivalent alternatives that perform the same tasks.

All relevant work, including code, diagrams, and circuit schematics, has been uploaded to the GitHub repository for easy reference and use.

The github repo : https://github.com/Ayboss/Smart-Insulin-Pump

5.2 RECOMMENDATIONS

1. Clinical Trials: Conduct extensive clinical trials to validate the accuracy and reliability of the NIR sensors in various real-world conditions and across diverse patient demographics.

2. User Experience Optimization: Gather and incorporate user feedback to further refine the mobile application's interface and features, ensuring maximum usability and engagement.

3. Integration with Healthcare Systems: Develop secure protocols for integrating the system with existing electronic health record systems to facilitate seamless data sharing with healthcare providers.

4. Miniaturization: Explore opportunities to further miniaturize the components, especially the insulin pump, to enhance comfort and wearability.

5. AI Integration: Incorporate machine learning algorithms to improve glucose prediction and insulin dosing over time, personalizing the system to individual patient needs.

6. Cost-effectiveness Analysis: Conduct a comprehensive cost-effectiveness analysis to determine the long-term economic benefits of the system for patients and healthcare systems.

By addressing these recommendations, the smart insulin pump system can be further enhanced, potentially revolutionizing diabetes management and improving the lives of millions affected by this chronic condition.

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